

Trust Region Subproblem: Exact Solution (1 + Bonus 1 pts)

Consider the following trust region subproblem in two dimensions, $v \in \mathbb{R}^2$:

$$\begin{aligned} &\text{minimize } m(v) = g^T v + \frac{1}{2} v^T B v \\ &\text{subject to: } \|v\| \leq \Delta \end{aligned}$$

as discussed in [Nocedal, Sec 4.3] and the slides. In every problem, given the inputs B, g and Δ , solve the task and plot the quadratic model and the trust region with the **tool**.

Problem 2:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with the Dogleg method.

Let's first see whether the Newton step is inside the trust region. The Newton step is $v^N = -B^{-1}g$. So first I will find B^{-1} using Gauss Jordan

$$[B|I] \rightarrow \left(\begin{array}{cc|cc} 4 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 1/4 & 0 \\ 0 & 1 & 0 & 1/2 \end{array} \right)$$

$$\text{So } B^{-1} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix} \quad \text{and} \quad v^N = - \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/4 \\ -1/2 \end{pmatrix}.$$

$$\|v^N\| = \sqrt{\frac{1^2}{4} + \frac{1^2}{2}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2} = 0.559$$

Newton step is outside the trust region because $\|v^N\| > \Delta = 0.5$

Let's compute $\|g\|$ to know in which dogleg case we are:

$$\|g\| = \sqrt{g^T g} = \sqrt{\frac{2}{(1 \ 1) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}} = \sqrt{\frac{2}{6}} = \frac{1}{\sqrt{3}} = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix}$$

$$\|v_g\| = \sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{3} = 0.471.$$

v_g is inside the trust-region so we are in the 3rd case.

$$v^* = v_g + (\gamma - 1)(v^N - v_g) \quad \text{with} \quad \|v^* + (\gamma - 1)(v^N - v_g)\| = \Delta^2$$

$$\begin{aligned} \text{So } v(\gamma) &= \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix} + (\gamma - 1) \begin{pmatrix} -1/4 + 1/3 \\ -1/2 + 1/3 \end{pmatrix} = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix} + \gamma \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} - \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} \\ &= \begin{pmatrix} -5/12 \\ -1/6 \end{pmatrix} + \gamma \begin{pmatrix} 1/12 \\ -1/6 \end{pmatrix} = \begin{pmatrix} -5/12 + \gamma/12 \\ -1/6 - \gamma/6 \end{pmatrix} \end{aligned}$$

We know that $\|v(\gamma)\|^2 = \Delta^2 \Rightarrow (-5/12 + \gamma/12)^2 + (\frac{1}{6} - \frac{\gamma}{6})^2 = \frac{1}{4} \Rightarrow$

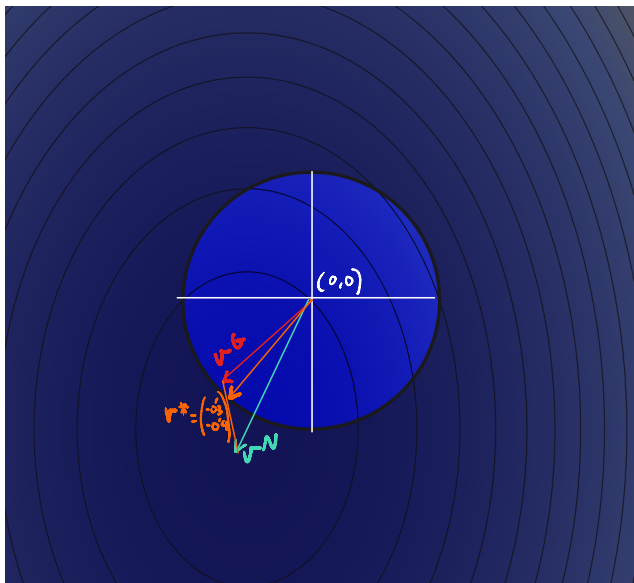
$$\Rightarrow \frac{25}{144} + \frac{-10\gamma}{144} + \frac{\gamma^2}{144} + \frac{1}{36} + \frac{2\gamma}{36} + \frac{\gamma^2}{36} = \frac{1}{4} \Rightarrow$$

$$\Rightarrow \frac{25}{144} - \frac{10\gamma}{144} + \frac{\gamma^2}{144} + \frac{4}{144} + \frac{8\gamma}{144} + \frac{4\gamma^2}{144} - \frac{36}{144} = 0 \Rightarrow$$

$$\Rightarrow 5\gamma^2 - 2\gamma - 7 = 0$$

$$\hookrightarrow \gamma = 1.4$$

$$v(1.4) = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix} \quad \|v(1.4)\| = 0.5$$



Problem 4:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

KKT Conditions:

$$\mathcal{L}(v, \lambda) = g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - \frac{1}{4})$$

1.- Primal constraints: $\|v\| \leq 0.5$

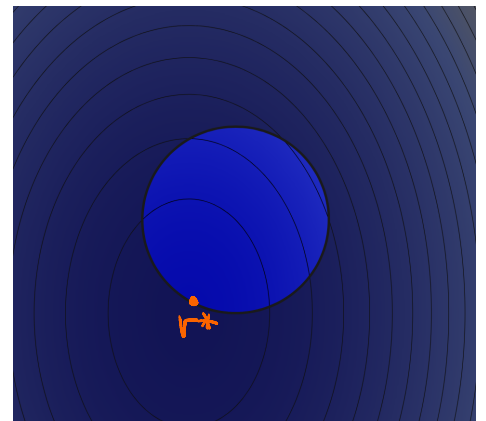
2.- Dual constraints: $\lambda_1 \geq 0$

3.- Complementary slackness: $\lambda(\|v\|^2 - \frac{1}{4}) = 0$

4.- Vanishing gradient of the Lagrangian:

$$\nabla_v \mathcal{L}(v, \lambda) = g + Bv + \lambda_1 v = 0$$

$$(B + \lambda_1 I)v = -g$$



Then for condition 4. to be true:

$$v(\lambda) = \begin{pmatrix} -\frac{1}{4+\lambda} \\ -\frac{1}{2+\lambda} \end{pmatrix}$$

As we know that the solution for the unconstrained problem lies outside the boundary, by complementary slackness v^* is in the boundary.

$$\|v(\lambda^*)\|^2 = \frac{1}{4} \Rightarrow \frac{1}{(4+\lambda^*)^2} + \frac{1}{(2+\lambda^*)^2} = \frac{1}{4} \quad \left\{ \text{using an online solver: } \lambda^* = 0.264 \right\}$$

$$\text{Then } v^* = v(\lambda^*) = \begin{pmatrix} -0.234 \\ -0.446 \end{pmatrix}$$

Problem 6: (bonus)

Consider now the the indefinite matrix

$$\tilde{B} = \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

KKT Conditions:

$$L(v, \lambda) = g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - 1)$$

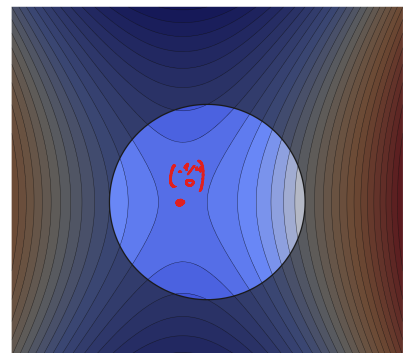
1.- Primal constraints: $\|v\| \leq 1$

2.- Dual constraints: $\lambda_1 \geq 0$

3.- Complementary slackness: $\lambda_1 (\|v\|^2 - 1) = 0$

4.- Vanishing gradient of the Lagrangian:

$$\begin{aligned} \nabla_v L(v, \lambda_1) &= g + Bv + \lambda_1 v = 0 \\ (B + \lambda_1 I)v &= -g \end{aligned}$$



Then for condition 4 to be true:

$$v_1 = -\frac{1}{4+\lambda}$$

$$(-2+\lambda)v_2 = 0 \Rightarrow v_2 = 0 \text{ or } \lambda = 2.$$

Let's try the case when $v_2 = 0$ and the solution is inside the boundary so $\lambda = 0$, by complementary slackness.

$$v(0) = \begin{pmatrix} -1/4 \\ 0 \end{pmatrix}.$$

So $\lambda = 0$, $v = \begin{pmatrix} -1/4 \\ 0 \end{pmatrix}$ satisfies KKT conditions although it is not the optimum due to the fact that B is not PSD.

If we assume the solution is on the boundary, $\lambda \neq 0$, and thus $v_2 \neq 0$.

In that case $v = \begin{pmatrix} -1/(4+\lambda) \\ 0 \end{pmatrix}$. $\|v\| = 1 \Rightarrow \frac{1}{4+\lambda} = 1 \Rightarrow \lambda = -3$.

But with $\lambda = -3$ $(B + \lambda I)$ is still not PSD.